SHIVAM AI TRADER

Features A to Z: everything the bot does, how it decides, and where it can be wrong

SHIVAM AI TRADER is a Telegram bot for crypto (Binance) and Indian markets (NSE stocks, indices, futures and options through Angel One). It scans markets with 7 rule-based strategies in BUY and SELL versions, draws charts with market structure, explains every condition, reads order books and option chains, builds futures and option trade plans, checks company fundamentals, and ranks market news by the sectors it can move. It runs on GitHub, Google Cloud or your own PC.

It does	It does not
Analyse with 7 strategies × BUY/SELL and explain each condition with real numbers	Place, change or cancel any order (there is no such code)
Draw support/resistance, trend lines, zones and structure on charts	Guarantee profit or predict prices
Show each strategy's history on that symbol (backtest in R)	Stream tick-by-tick prices
Read live order books, option chains, futures and OI build-up	Know your holdings or positions
Size positions from your capital and risk %	Replace your own judgement or a SEBI-registered adviser
Link each news story to sectors, stocks or crypto	Say whether news will push prices up or down

Risk
Everything the bot sends is for learning and paper trading, not financial advice. Paper-trade for at least 8 weeks with a journal before risking real money.

Section	Topic
1–2	How it works · where it runs · the Telegram menu and every command
3–6	Scanners · one-symbol analysis · the 7 strategies · structure and order book
7–11	Any NSE stock · option chain and Greeks · futures · OI build-up · fundamentals
12–14	News and geopolitics · automatic alerts · the AI's role
15–18	Data sources · safety design · self-test · limitations and risks
19	Glossary A to Z

1. How it works and where it runs

Step	What happens
1. Trigger	You send a command or tap a button, or an alert schedule fires
2. Data	Candles, order book, option chain, futures, fundamentals or news are fetched at that moment
3. Indicators	EMA 9/21/50/200, RSI 14, ATR 14, MACD, 20-candle volume average, relative volume
4. Structure	Swings, support/resistance, round numbers, trend lines, BOS/CHoCH, order blocks, fair-value gaps, demand/supply zones, candle patterns
5. Strategies	14 rule sets (7 BUY + 7 SELL) check the last completed candle
6. Backtests	Every strategy is replayed on the symbol's own history
7. Messages	Chart + trade card, strategy check, full explanation, buttons for next steps

Where it can run

Place	Replies	Alerts	Notes
GitHub Actions	Every 5 min (often later)	Yes	Free for public repositories; no server
Google Cloud Run	Seconds (webhook)	No (GitHub does them)	Free tier; images auto-cleaned
Google Cloud VM	Seconds (polling)	Yes	e2-micro; public IP costs ≈ $3–4/month
Your PC	Seconds (polling)	Yes	Only while the PC is on (start_bot.bat)

Only one place should answer messages and only one should send alerts; the setup guide explains the allowed combinations.

2. The Telegram menu and commands

Send `/start` for the main menu. Every screen ends with buttons for the next step and Menu, so you rarely need to type.

Menu button	What it opens
Crypto scan · NSE stocks scan	Numbered lists of BUY/SELL setups; tap a number
Any NSE stock	Type a symbol or company name, or tap a watchlist stock
Fundamentals	Company fundamentals for a stock you choose
Option chain · Futures	Choose NIFTY, BANKNIFTY, FINNIFTY, MIDCPNIFTY or type an F&O stock
News & geopolitics · OI build-up	Paged news · today's futures positioning
Strategies · ⚙ Settings · Help	Reference screens

Command	Result
/crypto · /stocks	Scanner lists (top 15 coins + 5 majors · NSE watchlist + NIFTY, BANKNIFTY)
/stock TATAMOTORS	Full analysis of any NSE stock
/search tata	Matching NSE symbols as buttons
/analyze ETH	One coin, stock or index (crypto first, then NSE)

Command	Result
/fundamentals INFY	Valuation, profitability, growth, debt, quarters, health check
/fno NIFTY	Option chain report, OI chart, Greeks, option trade idea
/futures RELIANCE	All expiries, basis, carry, OI build-up, chart, futures trade plan
/buildup	Long/short build-up lists across F&O futures
/news · /news india|world|crypto · /news SYMBOL	10 numbered stories per page
/strategies · /settings · /help · /start	Reference and menu
A number, e.g. 3	Item 3 of the list you received last (scanner or news)
Plain text: ETH · RELIANCE · tata motors	Opens that coin or stock; partial names show choices

3. Scanners

/crypto scans the 15 most-traded USDT pairs on Binance plus BTC, ETH, SOL, BNB and XRP (stablecoins and PAXG excluded) on 4-hour candles. /stocks scans the NSE watchlist (and NIFTY, BANKNIFTY with Angel One) on daily candles. The list is grouped and numbered:

Group	Meaning
● BUY setups / ● SELL setups	More strategies say TRADE on one side; shows how many of 7 agree
⚖ Mixed	BUY and SELL strategies both fire equally → no trade
● Forming	No TRADE yet, but some strategies have 2 of 3 conditions
○ No setup	Nothing close

Tap a number button or send the number. Numbers match the buttons, and the bot remembers the list.

4. One-symbol analysis (three messages)

Message	Contents
① Chart + trade card	BUY/SELL/forming/mixed/no setup · entry · take profit · stop-loss · reward:risk · position size for your capital · order-book reading · candle time and source
② Strategy check	The 7 strategies with ● ready, ● forming (2/3), ○ none, * new this candle, and the verdict
③ Full explanation	Every condition ✓/✗ with numbers · levels and size · each strategy's history · market structure and key levels · live volume and order book · optional AI summary
Buttons	Stocks: Option chain · Futures (F&O stocks) · Fundamentals · News · Refresh · ⬅ Back · Menu. Crypto: Crypto news · · ⬅ ·

Example trade card (simulated data)

```
 SOL/USDT · 4h
Price 97.20 USDT

● SELL SETUP · 1 of 7 strategies agree
──────────────────────

 Entry 97.20
 Take profit 92.54 (-4.80%)
 Stop-loss 99.54 (+2.40%)
⚖ Reward : Risk = 2 : 1
 Size 0.1906 SOL · max loss ≈ 0.50 USDT
Levels from: Trendline / Round-Number Breakout
History: 31 trades · win 29.0% · avg -0.21R
```

Note how honest the history line is: this setup's past average was −0.21R, i.e. it lost money on this symbol. That is exactly why the history is shown.



Chart with EMAs, volume, RSI, entry/stop/target, support/resistance with touch counts, round numbers, trend lines, zones and structure breaks (simulated prices).

5. The 7 strategies (each with BUY and SELL)

Each strategy has 3 conditions checked on the last **completed** candle. **TRADE** = 3/3, **WATCH** (forming) = 2/3, **NO** = 0–1. A **fresh** signal means the conditions became true on this candle; only fresh signals trigger alerts. Take profit = entry ± 2 × risk (2R, editable). Strategies are not used during the first 50 candles of history (indicator warm-up).

#	Strategy	BUY idea	SELL idea (mirror)	Stop (BUY; SELL mirrored)
1	EMA Trend Cross	EMA9 crosses above EMA21, price above EMA50	EMA9 crosses below EMA21, price below EMA50	1.5 ATR
2	20-Candle Breakout / Breakdown	Close above 20-candle high on strong volume	Close below 20-candle low on strong volume	Breakout level − 0.5 ATR or 1 ATR, whichever is lower
3	RSI Pullback in Trend	Uptrend; RSI dips below 40 and recovers	Downtrend; RSI rallies above 60 and turns down	5-candle low − 0.5 ATR (at least 0.5 ATR)
4	MACD Momentum	MACD crosses above signal while above zero	MACD crosses below signal while below zero	1.5 ATR
5	Support / Resistance Reaction	Dip into tested support or demand zone + engulfing or hammer	Rally into resistance or supply zone + engulfing or shooting star	Level − 0.3 ATR (at least 0.5 ATR)
6	Smart Money (BOS + OB/FVG)	After bullish BOS/CHoCH, retest of order block or fair-value gap, bullish close	Bearish mirror	Zone − 0.3 ATR (at least 0.5 ATR)
7	Trendline / Round-Number Break	Close above falling trend line or round number, volume, strong close	Close below rising trend line or round number, weak close	1.5 ATR

Verdict

More TRADE strategies on one side → BUY or SELL ("3 of 7 agree"). Equal numbers → MIXED (no trade). None but some forming → FORMING. The levels on the card come from the agreeing strategy with the best history on that symbol. A SELL means exit/avoid for cash holders; shorting needs futures or options (crypto: futures).

Backtest (history) rules

- Entry at the **next** candle's open (no look-ahead); slippage and fees on both sides (crypto 0.10%, stocks 0.15% per side, slippage 0.05%).
- If stop and target are both touched in one candle, the stop counts first; gaps fill at the open.
- Results in R (1R = the risk): trades, win rate, average R, total R, profit factor; fewer than 20 trades is flagged as weak evidence.
- History: crypto up to 3,000 four-hour candles (≈16 months); NSE ≈700 days (Angel One) or 2 years (Yahoo).

Position size

(capital × risk %) ÷ (distance to stop + round-trip costs), capped at your capital; whole shares for stocks. Defaults: ₹15,000 and 100 USDT at 0.5% risk (₹75 / 0.50 USDT per trade). Indices point to F&O.

6. Market structure and order book

Feature	How it's computed (causal: only candles up to that moment)
Swing highs/lows	Highest/lowest of 7 candles (3 each side), known 3 candles later; labelled HH/HL/LH/LL
Support / resistance	Swings of the last 200 candles clustered within 0.5 ATR; touch count = strength
Round numbers	Psychological levels scaled to price (e.g. 24,000 / 24,500; 100 / 50)
Trend lines	Through the last two lower highs (falling) or higher lows (rising)
BOS / CHoCH	Close beyond the last swing; same direction = BOS, reversal = CHoCH
Order blocks	Last opposite candle before a structure break; removed when closed through
Fair-value gaps	3-candle gaps ≥ 0.1 ATR; removed when filled
Demand / supply zones	Base candle followed by a move of ≥ 1.5 ATR; removed when closed through
Candle patterns	Engulfing (bull/bear), hammer, shooting star

Live volume and order book (context, not a rule)

- **Crypto (Binance):** bids vs asks within ±1%, biggest walls, spread, aggressive-buying % of the last candle, volume pace of the forming candle.
- **NSE (Angel One):** total pending buy vs sell quantity, biggest walls in the best 5 levels, spread, today's volume pace vs average. Indices use the near-month future's book.
- Reading: **buyers stronger** (pressure ≥ 1.2× or aggressive buying ≥ 55%), **sellers stronger** (≤ 0.83× or ≤ 45%), otherwise **mixed**. Order books change in seconds and can be spoofed, so they never decide a signal.

7. Any NSE stock

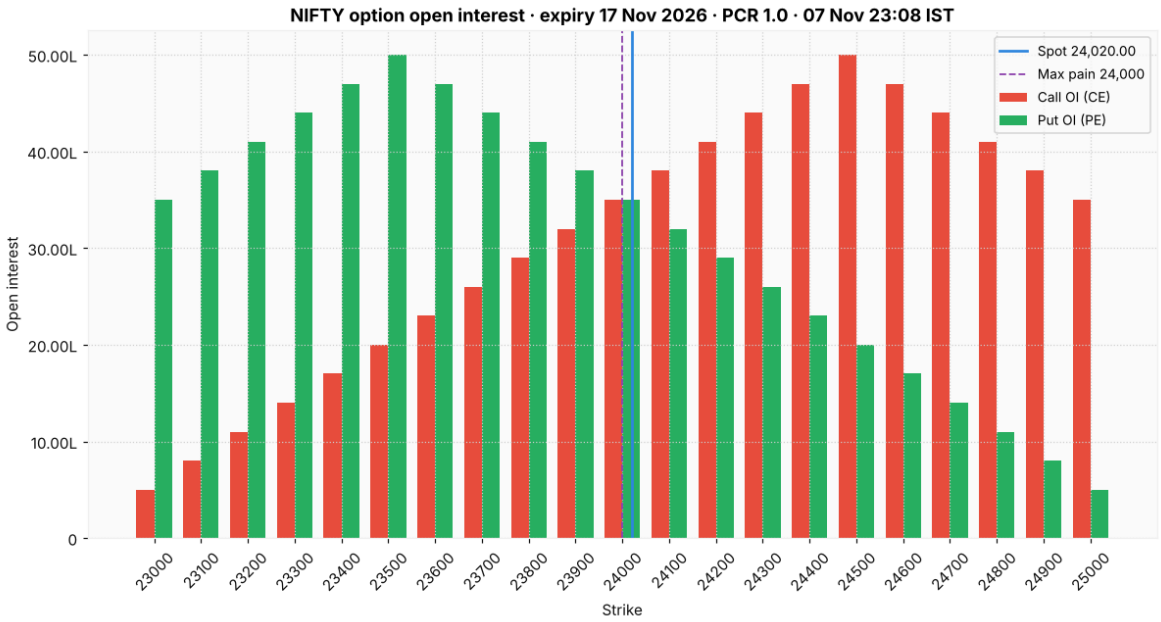
- Exact symbols open directly: `RELIANCE`, `M&M`, `BAJAJ-AUTO`.
- Partial symbols or names show choices: `tata` → TATAMOTORS, TATASTEEL, TATAPOWER … as buttons.
- Company names: `tata motors` → TATAMOTORS (Angel One's instrument list, then Yahoo's name search).
- After the Any NSE stock, Fundamentals, Option chain or Futures button, the next thing you type is used for that screen (for 15 minutes).
- Option chain and futures buttons appear only for stocks that have F&O contracts.

8. Option chain, Greeks and option trade idea (/fno)

Part	Contents
OI chart	Call vs put open interest by strike, spot line, max-pain line (photo)
Header	Spot and change, nearest expiry, lot size; near-month future price, basis, OI, notional
Chain reading	PCR (≥ 1.05 put-heavy, ≤ 0.95 call-heavy, else balanced), highest call OI (watched as resistance), highest put OI (support), max pain and distance from spot
Greeks	For the at-the-money call and put: **IV, delta, theta**, plus premium per lot vs your capital
Chain table	CE OI, CE LTP, strike, PE LTP, PE OI for strikes around ATM (* marks ATM)
Option trade idea	From the chart verdict: BUY → ATM call, SELL → ATM put. Premium × lot = cost; premium at take profit and at stop estimated with delta; worst case (whole premium); time decay per lot per day; warning near expiry

Greeks

Greeks coverage: Angel One's option-Greeks service returns delta, gamma, theta, vega and IV for the nearest expiry during market hours. The bot currently shows **delta, theta and IV** for the ATM call and put and uses delta and theta in the trade idea. **Gamma and vega are not shown**, and the chain table has no per-strike Greeks. Outside 09:15–15:30 IST the Greeks are empty and the report says so.



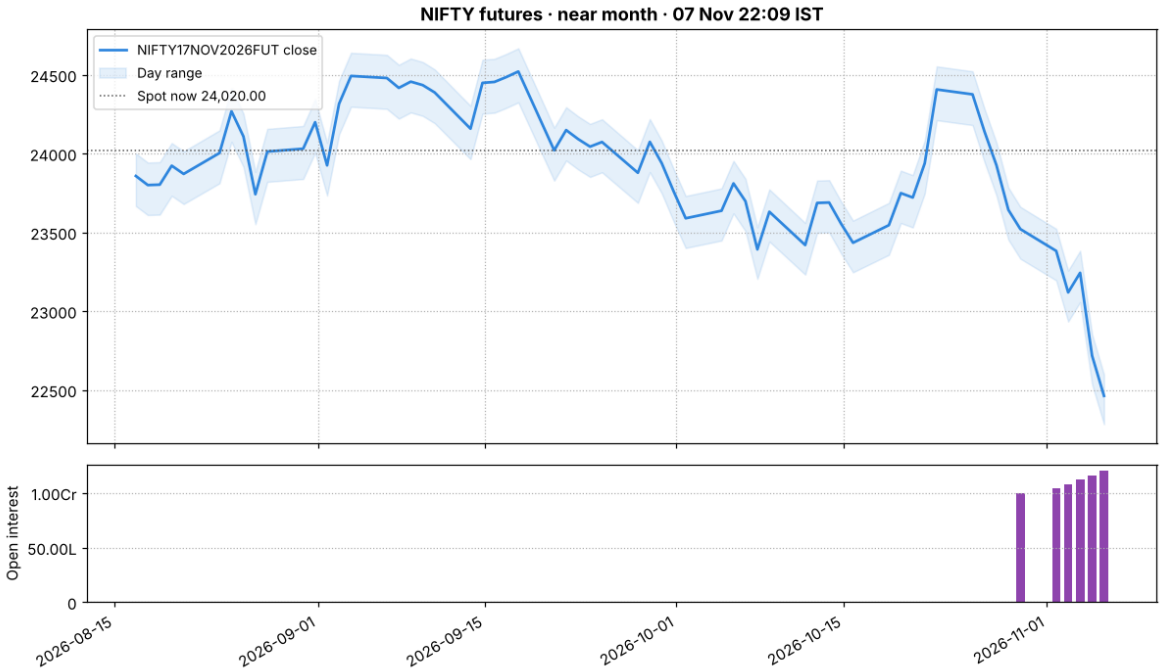
Open-interest chart from the F&O report (simulated data).

9. Futures report (/futures)

Part	Contents
Every expiry	Near, next and far month: price and change, day range, basis vs spot (points and %), annualised cost of carry, OI, volume, lot size, notional value of 1 lot
Reading	Premium/discount explained; next − near spread (roll cost); OI build-up; volume ÷ OI
Chart	Near-month daily close and range with spot line, and daily open interest bars (when Angel One returns OI history)
Futures trade plan	Chart verdict shifted into futures prices (spot level + basis): entry, take profit, stop, risk and reward per lot, % of capital at risk vs your limit, with a warning when 1 lot is too big

OI build-up	Price	OI	Usual reading
● Long Built Up	↑	↑	New buying positions (bullish)
● Short Built Up	↓	↑	New selling positions (bearish)
● Short Covering	↑	↓	Sellers exiting; rise may be short-lived
● Long Unwinding	↓	↓	Buyers exiting; weakness, not fresh shorts

The label comes from Angel One's build-up lists (top contracts of the day); if the contract isn't in them, the bot classifies it from price change and day-on-day OI change when OI history is available.



Futures chart: near-month close and range, spot line, open-interest bars (simulated).

10. OI build-up (/buildup)

The top 5 near-month contracts in each of Angel One's four lists (Long Built Up, Short Built Up, Short Covering, Long Unwinding) with price and change, plus buttons to open each one's futures report. Empty outside market hours.

11. Fundamentals (/fundamentals)

From Yahoo Finance for SYMBOL.NS: company, sector and industry; market cap (₹ crore / lakh crore); price and position in the 52-week range; valuation (P/E, forward P/E, P/B, EPS, book value); profitability (ROE, ROA, operating and net margin); growth vs last year's quarter; balance sheet (debt/equity, current ratio, cash, debt); dividend yield and payout; insider/promoter and institutional holding, beta; last 4 quarters' revenue and net profit.

Health check	✓	⚠	✗
Return on equity	≥ 15%	10–15%	< 10%
Debt / equity (not for banks)	≤ 0.5	0.5–1.5	> 1.5
Net profit margin	≥ 10%	5–10%	< 5%
Revenue growth	≥ 10%	0–10%	< 0%
Earnings growth	≥ 10%	0–10%	< 0%
P/E	0–30	30–60	> 60 or loss-making
Current ratio (not for banks)	≥ 1.2	1.0–1.2	< 1.0
Free cash flow (not for banks)	positive	–	negative

Verdict: **Strong** (≥ 70% pass and none fail), **Weak** (half or more fail), otherwise **Mixed**. Units are handled carefully (Yahoo's debt/equity is a percentage; dividend yield is computed from the dividend rate). Yahoo is unofficial and can be late or incomplete.

12. News and geopolitics (/news)

Tab	Sources (public RSS feeds)
India	Economic Times (markets, stocks, economy), Mint markets, Google News (Sensex, Nifty, RBI, SEBI)
World	Google News (geopolitics, war, sanctions, tariffs, OPEC, Fed; crude, dollar, yields), Economic Times international, BBC World
Crypto	CoinDesk, Cointelegraph, Economic Times crypto
All	Everything above, merged
/news SYMBOL	Google News for that stock, last 7 days

How a page looks

```
 MARKET NEWS ·  All · page 1/4
1. Brent crude jumps 4% as Iran tensions flare — links to oil & gas (ONGC,
   Reliance), OMCs (BPCL, IOC), airlines, paints and the rupee  1m ago
2. Fed signals two more rate cuts as US inflation cools — links to foreign
   flows into NIFTY, IT stocks and crypto (BTC, ETH)  1h ago
8. Tata Motors Q2 profit falls 30% on JLR weakness — links to TATAMOTORS directly
[1][2][3][4][5] [6][7][8][9][10]   [◄ Prev] [· 1/4 ·] [Next ►]
[ All] [ India] [ World] [ Crypto]   [ Refresh] [ Menu]
```

- **10 stories per page**, each with a **serial number**, one sentence, and "links to ..." naming the sector, stock or crypto. Next/Prev flip the page in place like website tabs.
- **Send the number** (e.g. 12) or tap it for the detail: headline, source and time, what happened (feed summary or the article's own description), every market link with a one-line "why", stocks and crypto to watch as buttons (open their analysis), Read full article, Next story, Back to list, and an optional AI take.
- **Ranking:** strongest market theme, number of themes, named companies and freshness (3h, 8h, 24h); stories older than 48 hours dropped; world stories with no market angle dropped; near-duplicates removed; up to 50 stories.

Theme (examples of words)	Links to
Crude, Brent, OPEC	Oil & gas (ONGC, Reliance), OMCs (BPCL, IOC), airlines, paints, rupee
RBI, repo rate, MPC	Banks & NBFCs, BANKNIFTY, realty, autos
Fed, FOMC, US yields, US inflation	Foreign flows into NIFTY, IT stocks, crypto
War, missiles, Iran, Israel, Russia, Taiwan...	Risk sentiment: India VIX, crude, gold, defence (HAL, BEL), crypto
Tariffs, sanctions, trade deal	Exporters (IT, pharma, textiles, auto parts), metals
Bitcoin, Ethereum, ETF, stablecoins	Crypto market
Rupee, dollar index, forex reserves	IT/pharma exporters vs importers
FII/FPI/DII flows	NIFTY and BANKNIFTY direction
Gold, silver	Titan, Muthoot Finance, gold ETFs
GDP, CPI, budget, GST	Broad market, banks, consumer stocks
SEBI, F&O rules, lot size	F&O traders, brokers and exchanges (BSE, Angel One, CDSL)
IT, pharma, auto, metals, banks, monsoon, telecom, power, defence, realty, infra, IPOs, Wall Street	The matching sector and its leading stocks
≈70 company names (Reliance, Infosys, HDFC Bank, Tata Motors...)	That stock directly

> **Note**
> Links come from fixed keyword rules, so they are consistent and never invented. They say who usually reacts, not which way prices will move. The AI take (if Gemini is set) uses only the headline and summary and never says buy or sell.

13. Automatic alerts

What	When (IST)	Sends
Crypto	≈7 min after each 4h candle: 01:37, 05:37, 09:37, 13:37, 17:37, 21:37	Coins with a fresh signal on the winning side (max 5; more are listed by name)
Stocks and indices	16:10 on weekdays	Watchlist names with a fresh signal
F&O snapshots	09:45 and 14:30 on weekdays	NIFTY and BANKNIFTY option-chain reports

Alerts use the same three-message format, headed NEW SIGNAL. GitHub can start schedules late; a PC or VM in local mode runs the same times (within 30 minutes). NSE holidays aren't detected.

14. The AI's role (optional Gemini)

- Python decides everything: indicators, levels, signals, sizes, links.
- Gemini only rewrites computed facts into a short plain-English summary (≤ 110 words) and, for a news story, explains which sectors it may affect (≤ 70 words). It is told not to add numbers, predictions or buy/sell advice.
- If the key or model is missing or fails, messages are sent without the AI part.

15. Data sources

Data	Source	Live?
Crypto candles, order book, top coins	Binance public market data (Kraken fallback for candles)	Yes
NSE stocks, indices, depth	Angel One SmartAPI (Yahoo fallback, delayed)	Yes with Angel One
Option chain, Greeks, futures, OI build-up, OI history	Angel One SmartAPI	Yes; Greeks in market hours
Instrument list (symbol search)	Angel One public scrip master	Daily file
Fundamentals, company-name search	Yahoo Finance via yfinance (unofficial)	Delayed
News	Public RSS feeds (Economic Times, Mint, Google News, BBC, CoinDesk, Cointelegraph)	Minutes
AI text	Google Gemini with your key and model	–

16. Safety design

- No order code at all; only read-only Angel One calls (login, instruments, candles, quotes, Greeks, OI). SEBI's 2026 algo rules apply to API orders, which the bot never sends.
- Answers only your TELEGRAM_CHAT_ID; button taps from other chats are ignored.

- Cloud Run endpoint checks Telegram's secret header (derived from your token) and ignores repeats.
- Secrets live only in GitHub Secrets, Cloud Run settings or your private .env file.
- Signals on completed candles only; structure computed causally; backtests enter at the next open (automated no-look-ahead tests).
- 84 automated tests run on every GitHub upload, with fake Angel One, Telegram and news data.
- Google Cloud images are cleaned after each deploy and daily, to stay in the free 0.5 GB.

17. Self-test

GitHub → Actions → Self-test (or `python -m bot.main selftest`). The Telegram report checks crypto data and order book, stock data, Angel One login, instrument list, NIFTY option chain, OI, Greeks, futures (expiries, OI history, build-up), news feeds working, fundamentals, Gemini, and lists the field names Angel One returned.

18. Limitations and risks

Area	What to know
No guaranteed profit	Textbook strategies; histories are in-sample and often small; many signals can be one correlated bet; a SELL/BUY verdict with poor history should be ignored
F&O	Option premiums can go to zero; time decay and IV crush; futures losses can exceed margin. SEBI's study found about 93% of individual F&O traders lost money in FY22–FY24
Option estimates	Premium at target/stop uses delta only (ignores theta, gamma, IV changes)
Futures risk per lot	One NIFTY lot can risk ~100% of a ₹15,000 capital; the plan warns you
Crypto	24×7 volatility, exchange risk; India taxes gains at 30% with 1% TDS — ask a CA
Timing	Completed candles plus schedule delays: alerts can be minutes old; check live prices
Data quality	Yahoo is unofficial; Greeks/depth only in market hours; OI history format is undocumented by Angel One and skipped if unreadable; news feeds can fail or change
News links	Keyword-based: a headline can mention a word without real impact
Service	GitHub may restrict non-software use; schedules pause after 60 days without changes; Cloud Run needs an upgraded billing account after the trial; a VM's IP costs money
Security	The four Angel One secrets can log in to your account; protect them
Behaviour	Alert fatigue and overtrading are real risks; keep a journal and follow your limits

19. Glossary A to Z

Term	Meaning
ATM / ATR	At-the-money strike (closest to spot) / Average True Range, typical candle size (14)
Basis / carry	Futures − spot; carry = basis as a yearly % of spot
BOS / CHoCH	Break of structure (trend continues) / change of character (trend may turn)
Delta / theta / IV	Premium change per ₹1 move / premium lost per day / implied volatility
Demand / supply zone	Area where a strong up / down move started
EMA	Exponential moving average (9, 21, 50, 200 candles)
FVG	Fair-value gap: a 3-candle price gap the market often revisits
Gamma / vega	How delta changes / premium change per 1% IV (not shown by the bot)
Lot	Fixed F&O contract size set by the exchange
MACD	Momentum indicator: EMA12 − EMA26 vs its 9-period signal
Max pain	Strike where option buyers' total payoff at expiry would be smallest
OI / OI build-up	Open interest (open contracts) / how price and OI moved together
Order block	Last opposite candle before a strong structure break
PCR	Put-call ratio of open interest
R (risk unit)	Distance from entry to stop; 2R target = twice that distance
RSI	Relative Strength Index (14), 0–100 momentum gauge
Round number	Psychological level like 24,000 or 100,000
Slippage	Difference between expected and actual fill price
Support / resistance	Price areas where buying / selling repeatedly appeared
TOTP	Time-based one-time password used for Angel One login
Webhook / polling	Telegram pushes messages to a server / the bot asks Telegram for messages

Nothing in this document or in the bot's messages is financial advice.